

Inspecting failures and anomalies

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Running the primary experiment

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Power and resource tradeoffs

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy and data governance

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Inspecting failures and anomalies. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled inspecting failures and anomalies evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of inspecting failures and anomalies. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

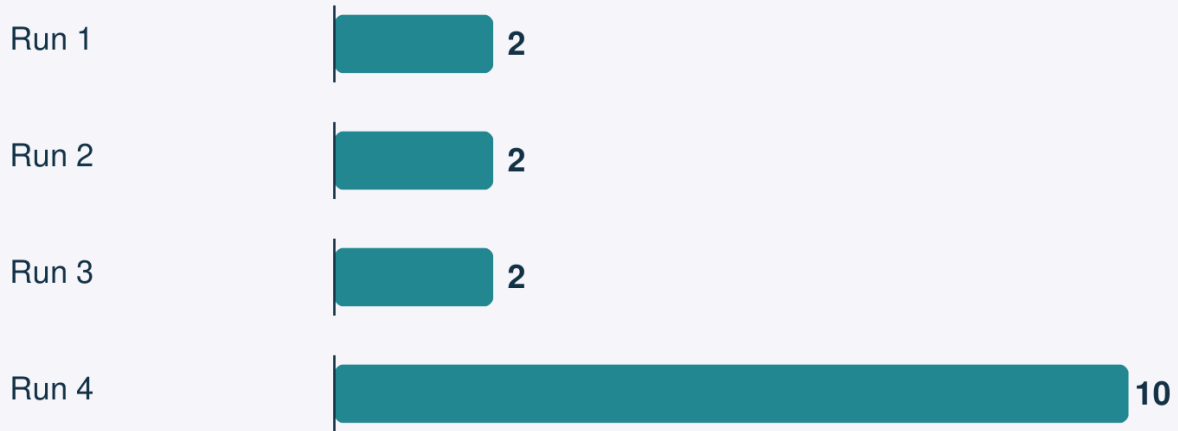
12. Pictorial case practice: Inspecting failures and anomalies

Inspect an anomaly without erasing it. A timing study records four runs and notices one unusually long duration.

Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run. The plan includes timeouts in reporting.

All four recorded durations

seconds; run 4 includes timeout



Bar length shows the supplied value; the marked baseline is zero.

Inspecting failures and anomalies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep the raw record

The fourth duration remains 10 seconds.



2. Compute summaries

Mean is $16/4=4$ seconds; median is 2 seconds.



3. Investigate the event

Report the timeout evidence and distinguish it from an untested explanation.

CHANGE ONE THING

Delete the 10-second record only because it looks bad.

Explain the changed result using the matching lesson.

Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inspect an anomaly without erasing it. A timing study records four runs and notices one unusually long duration.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report the timeout evidence and distinguish it from an untested explanation.

1. Investigate the event
2. Keep the raw record
3. Compute summaries

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Investigate the event Report the timeout evidence and distinguish it from an untested explanation. An anomaly invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Practice 2: explain the deciding evidence

Changed-case practice: Delete the 10-second record only because it looks bad. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The mean becomes 2 seconds but the selective report violates this plan.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Repair a missing building block

Review Running the primary experiment (ai-7/25). Goal: Explain the mechanism and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the mechanism and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the mechanism and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key. Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

